

Knowledge Engineering Guide

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ISO Standards for guiding our Knowledge Engineering:

ISO (International Organization for Standardization) has several standards that are relevant to the construction and maintenance of taxonomies and ontologies. Here's a list of key ISO standards specifically associated with taxonomies, ontologies, and related concepts:

ISO 25964:

- Part 1: **ISO 25964-1** - Thesauri and interoperability with other vocabularies
– Part 1: Thesauri for information retrieval.
- Part 2: **ISO 25964-2** - Thesauri and interoperability with other vocabularies
– Part 2: Interoperability with other vocabularies.

ISO 5964 - Guidelines for the establishment and development of multilingual thesauri. (Note: This standard has been withdrawn and replaced by ISO 25964-1).

ISO/IEC 13250 - Topic Maps:

- The standard specifies a model and a syntax for topic maps, which can be used to represent the structure of knowledge.

ISO 19763 - Metamodel framework for interoperability (MFI):

- This is a series of standards providing a framework for building ontologies for various domains and ensuring their interoperability.

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ISO 704 - Terminology work – Principles and methods:

- While not strictly about taxonomies or ontologies, this standard sets forth the principles and methods for creating terms, crucial for creating standardized vocabularies.

ISO 1087-1 - Terminology work – Vocabulary – Part 1: Theory and application:

- This part focuses on the theory of terminology, essential for building consistent and well-structured taxonomies.

ISO 21127 - Information and documentation – A reference ontology for the interchange of cultural heritage information:

- This standard specifies a reference ontology for the purposes of interchange of cultural heritage information.

ISO 80000-1 - Quantities and units – Part 1: General:

- While not strictly a taxonomy or ontology standard, it does provide standard naming and symbols for quantities and units, essential for some technical taxonomies.

ISO 30300 - Information and documentation - Management systems for records - Fundamentals and vocabulary:

- This provides a foundation for building taxonomies and ontologies in the context of record management.

ISO 19763 - Metamodel framework for interoperability (MFI)

Summary: ISO 19763 is a series of standards that provide a framework for metamodeling and ensuring interoperability among models. The primary objective of this series is to standardize the structure of metamodels related to various standards and to ensure that different models can be understood in relation to one another. This facilitates data exchange and interoperability between different systems or software tools.

Example: Applying interoperability between Telcos (5 common services)

1. Service Interoperability:

- Consider two telecom companies, Company A and Company B. Both use different models for representing their services and offerings. Company A might use a proprietary system for network management, while Company B might use a different system.
- ISO 19763 can help create a metamodel that provides an overarching framework, allowing both companies to understand the services in the context of the metamodel. This can be essential, for instance, when these companies are trying to achieve network interoperability or when a merger between the two companies takes place.

2. Customer Data Exchange:

- Telecom companies frequently need to exchange customer data, especially when a customer switches providers. One provider might model customer data differently from another.
- Utilizing the ISO 19763 framework, a common metamodel can be created for customer data. This metamodel would define the structure and semantics of customer data, ensuring that when Company A sends customer data to Company B, Company B can understand and integrate that data without ambiguity.

3. Equipment and Infrastructure Modeling:

- Different telecom operators might use different models to represent their network infrastructures, including towers, relay stations, and other equipment.
- A metamodel based on ISO 19763 can define the standard representation of such equipment and infrastructure. For instance, what constitutes a 'relay station' or how to represent the bandwidth

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capabilities of a particular tower. This becomes crucial when planning shared infrastructure or co-using towers and equipment.

4. **Billing and Usage Patterns:**

- Telecom companies have complex billing models based on user behavior, subscription packages, and other factors.
- Through ISO 19763, a standardized metamodel can be established that allows different companies to understand and analyze usage patterns and billing data in a consistent manner. For example, understanding how 'peak usage hours' are defined and modeled by different providers.

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Service Interoperability Ontology for Teleco:

Note: Domain Modeling and Ontology Design is a comprehensive process, involving domain experts (Bus SMEs) and Knowledge Engineer(s) working collaboratively using iterative feedback loops, and rigorous testing to ensure the design meets the specific needs and accurately represents the domain.

- **Concepts (Classes):**
 - **Service:** Represents any service provided by a Telco.
 - Subclasses: **VoiceService**, **DataService**, **MessagingService**, **RoamingService**, etc.
 - **Network:** Represents the telecom's physical and virtual infrastructure.
 - Subclasses: **2G**, **3G**, **4G**, **5G**, **Wi-Fi**, **Satellite**, etc.
 - **Device:** Represents devices using the services.
 - Subclasses: **MobilePhone**, **Router**, **Tablet**, **IoTDevice**, etc.
 - **Subscription:** Represents customer subscriptions to services.
 - Subclasses: **Prepaid**, **Postpaid**, **Contract**, **PayAsYouGo**, etc.
 - **Customer:** Represents the Consumer of services.
 - **Service_Provider:** Represents one or more Telco service providers to support interoperability to provide a seamless UX.
- **Relationships (Properties):**
 - **usesService:** Relates a **Device** or **Customer** to a **Service**.
 - **supportsService:** Relates a **Network** to a **Service** indicating which services are supported by which network types.
 - **subscribedTo:** Relates a **Customer** to a **Subscription**.
 - **providesService:** Relates a **ServiceProvider** to a **Service**, indicating which services are offered by which provider.
 - **interoperableWith:** Relates one **Service** or **Network** to another, indicating compatibility or interoperability.
 - **requiresDevice:** Relates a **Service** to a **Device**, indicating what devices are necessary for a particular service.

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- **Attributes (Data Properties):**

- **Service:**
 1. **serviceName**
 2. **serviceID**
 3. **serviceDescription**
 4. **serviceCost**
- **Network:**
 1. **networkType**
 2. **networkCoverage**
 3. **networkSpeed**
- **Device:**
 1. **deviceName**
 2. **deviceOS**
 3. **deviceCompatibility**
- **Subscription:**
 1. **subscriptionType**
 2. **subscriptionCost**
 3. **subscriptionDuration**
- **Customer:**
 1. **customerID**
 2. **customerName**
 3. **customerAddress**

- **Instances (Individuals):**

- **Services:** VoiceCall, MobileInternet, InternationalRoaming, etc.
- **Networks:** 4GLTE, 5GNR, HomeWi-Fi, etc.
- **Devices:** iPhone13, SamsungGalaxyS21, LinksysRouter, etc.
- **Subscriptions:** UnlimitedDataPlan, FamilyPlan, etc.
- **ServiceProviders:** TelecomCompanyA, TelecomCompanyB, etc.

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ISO/IEC 21838: Information technology — Top-level ontologies (TLO).

An upper or top-level ontology provides a high-level framework that can be used to structure domain-specific ontologies, ensuring consistency and interoperability among them. It typically deals with very general concepts like "entity", "relationship", "property", "time", and "event", which are universal across domains. The ISO/IEC 21838 standard aims to set criteria and best practices for the design and documentation of such ontologies.

ISO/IEC 21838 consists of multiple parts:

1. **ISO/IEC 21838-1:** This part provides an overview, including the scope, definitions, and basic principles for top-level ontologies.
2. **ISO/IEC 21838-2 and subsequent parts:** These parts specify individual top-level ontologies.

Companies and organizations often use upper ontologies (also known as top-level ontologies or foundational ontologies) to map relationships between Knowledge Organization Systems (KOS), which include taxonomies, thesauri, classification schemes, and other ontologies.

Here's why and how upper ontologies are used in this context:

- **Standardization:** Upper ontologies provide a standardized set of general concepts that can be used across various domains. By adhering to these concepts, different KOS can achieve a level of uniformity.
- **Interoperability:** When KOS from different domains or developed by different stakeholders need to interact or share information, upper ontologies can serve as a common reference point. This ensures that similar concepts from different KOS are understood to be the same, and relationships are correctly interpreted.
- **Semantic Integration:** In scenarios where multiple databases or information systems are merged, different KOS might be in place. An upper ontology can help in mapping similar concepts from these systems to a single unified concept, aiding in the integration process.
- **Enhanced Querying:** With the relationships between different KOS mapped using an upper ontology, it becomes possible to run more comprehensive and semantically enriched queries across systems.

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- **Consistency:** Upper ontologies ensure that the same concepts and relationships are represented consistently across various KOS. This is especially useful for large organizations or consortiums where multiple teams might develop their own KOS.
- **Extensibility:** Since upper ontologies deal with general concepts, they can be a foundation upon which more specific domain ontologies are built. This allows for extensibility as new domains or subdomains emerge.

[Reference Taxonomy 2.0 Proposal for introducing an upper ontology based on BFO \(basic formal ontology\)](#)

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Applying ISO/IEC 21838

Applying a TLO to map domain-specific Knowledge Organization Systems (KOS):

Scenario:

Suppose a multinational company has different departments handling finance, human resources, and product management, each with its own KOS.

An upper ontology could help in mapping concepts across these systems. For instance, the term "employee" in the HR system, "account holder" in the finance system, and "product manager" in the product management system might all map to a general "person" concept in the upper ontology.

While upper ontologies offer many benefits, their implementation can be challenging due to the need for extensive collaboration, expertise in ontology design, and sometimes the sheer complexity and breadth of information to be covered.

1. Identification of Core Concepts:

- Using a TLO approach, identify the high-level, domain-agnostic concepts relevant to the telecom sector. For instance, core concepts might include **Entity, Process, Relationship, Time, Location, Event, and Resource**.

2. Mapping Domain-Specific Concepts:

- For each domain within the telecom sector, align specific concepts with the core concepts identified in the TLO. For instance:
 - **Network Domain:**
 - 2G, 3G, 4G, 5G -> **Resource**
 - Network Failures -> **Event**
 - Network Coverage -> **Location-Based Attribute**
 - **Customer Services Domain:**
 - Subscriber -> **Entity**
 - Billing -> **Process**
 - Subscription Renewal -> **Event**

3. Establishing Relationships:

- Using the TLO, define the relationships between various concepts. For example, a **Subscriber (Entity)** might have a **Billing (Process)** event every month (**Time**).

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4. Integrate Multiple KOS:

- Suppose there are different KOS for network management, customer relationship management (CRM), and supply chain within the telecom company. The TLO can be utilized as a unifying framework, mapping concepts from each KOS to the TLO and ensuring a level of interoperability among these systems.

5. Enhancing Data Retrieval:

- By aligning all domain KOS with the TLO, it becomes easier to execute cross-domain queries. For example, an executive might want to see the impact of **Network Failures** (**Event** from Network Domain) on **Billing** (**Process** from Customer Services Domain). Since both concepts are mapped to the TLO, it's easier to understand and retrieve this interconnected data.

6. Semantic Integration & Interoperability:

- In cases where the telecom company uses different software solutions or databases for each domain, a TLO approach, in line with ISO/IEC 21838, can aid in semantic integration. By ensuring every system aligns with the TLO, it ensures semantic interoperability and data consistency.

7. Supporting Evolving Knowledge:

- As the telecom sector evolves, new concepts and technologies will emerge. Using a TLO approach means these new concepts can be mapped to the existing top-level ontology, ensuring continuity and reducing the complexity of incorporating new knowledge.